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Week	Week 7

WEEK-7 TASK

Task Name

Unsupervised Financial Anomaly Detection using Isolation Forest

Abstract

Isolation Forest is an unsupervised algorithm used to detect unusual financial transactions. It identifies anomalies by randomly separating data and calculating anomaly scores.

Introduction

Financial data may contain unusual transactions that indicate fraud or errors. Isolation Forest helps detect these anomalies without labelled data by using random isolation trees.

Objective:

Build an Isolation Forest without using a ready-made anomaly detection library. The algorithm should identify unusual transactions by isolating them through random binary splits.

Main Steps

1. Prepare transaction data

- Transaction amount
- Transaction frequency
- Account balance
- Transaction time
- Other financial attributes

2. Build an Isolation Tree

- Randomly select a feature.
- Randomly select a split value.
- Divide the data into two groups.
- Repeat recursively until the data is isolated or maximum tree height is reached.

3. Create an Isolation Forest

- Build many isolation trees.
- Each tree uses a random sample of transactions.
- Calculate the path length of each transaction in every tree.

4. Calculate Anomaly Score

The basic score is:

$$s(x, n) = 2^{-\frac{E(h(x))}{c(n)}}$$

Where:

- x = transaction
- $E(h(x))$ = average path length across all trees
- $c(n)$ = average path length of an unsuccessful search in a Binary Search Tree
- n = sample size

Interpretation:

- Score close to 1 → likely anomaly
- Score around 0.5 → normal/uncertain
- Lower score → more likely normal

Expected Workflow

Financial Transaction Data



Data Preprocessing



Random Sub-sampling



Build Isolation Trees



Calculate Path Lengths



Average Path Length



Anomaly Score



Normal Transactions / Anomalies

Conclusion

Isolation Forest provides an effective way to detect unusual financial transactions without requiring labelled data. By randomly partitioning transactions and measuring their isolation path lengths, anomalous transactions can be identified efficiently. Implementing the algorithm from first principles helps understand its working, mathematical scoring, and practical use in financial fraud and anomaly detection.